code listing shows the implementation of the corresponding function in the server.js JavaScript file, acting as a web server.

Listing 7.2 Retrieving all the image locations from the S3 bucket

```
const bucket = process.argv[2];                    ①

async function listImages(response) {
  try {
    let data = await s3.listObjects({              ②
      Bucket: bucket                               ③
    }).promise();
    let stream = mu.compileAndRender(              ④
      'index.xhtml',
      {
        Objects: data.Contents,
        Bucket: bucket
      }
    );
    stream.pipe(response);                         ⑤
  } catch (err) {                                  ⑥
    console.error(err);
    response.status(500);
    response.send('Internal server error.');
  }
}
```

① Reads the bucket name from the process arguments

② Lists the objects stored in the bucket

③ The bucket name is the only required parameter.

④ Renders an HTML page based on the list of objects

⑤ Streams the response

⑥ Handles potential errors

Listing the objects returns the names of all the images from the bucket, but the list doesn't include the image content. During the uploading process, the access rights to the images are set to public read. This means anyone can download the images with the bucket name and a random key. The following listing shows an excerpt of the index.xhtml template, which is rendered on request. The `Objects` variable contains all the objects from the bucket.

Listing 7.3 Template to render the data as HTML

```
[...]
<h2>Images</h2>
{{#Objects}}          ①
  <p><img src=        ②
```